Program : **Diploma in Renewable Energy**	
Course Code : **3293**	Course Title: **Renewable Energy Engineering I**
Semester : **3**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester:**60**

Course Objectives:

- To understand solar radiation.
- To understand methods of utilization of solar thermal energy.
- To understand about solar PV cells and design of solar panels.
- To select components for PV installation.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Renewable Energy Technologies Charging methods and interconnections of batteries.		Basic Electrical Engineering & Renewable Technologies	**2**

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Classify various methods of harnessing solar energy	14	Understanding
CO2	Explain the various technologies for solar heating systems	15	Understanding
CO3	Identify different interconnections of Solar Photovoltaic (SPV) modules.	15	Applying

CO4	Identify various components in Balance Of Systems (BOS) for PV installation.	14	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				3		
CO2	2				3		
CO3	3				3		
CO4	3				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Classify various methods of harnessing solar energy**		
M1.01	Outline the scope of solar energy and discuss its potential.	2	Understanding
M1.02	Summarize the characteristicsof solar radiation collecting systems.	4	Understanding
M1.03	Illustrate the construction and operation of various types of solar collectors.	5	Understanding
M1.04	Explain the basis of Solar thermal storage systems.	3	Understanding

Contents:

Solar energy -The sun as source of energy-scope and potential of solar energy-reasons to promote solar energy-advantages and disadvantages of solar energy - applications of solar energy(listing only)-basic concepts of solar energy conversion techniques (solar thermodynamics, solar photovoltaic and greenhouse effect).

Solar Thermal energy collectors -Classification based on temperature (low, medium and high temperature) -characteristic features of a collector system(listing only) -factors affecting collector system efficiency (shadow factor, cosine loss factor and reflective loss factor).

Solar collectors-Types-diagrams and principle of operation of various collectors - concentrating type (parabolic trough, parabolic dish ,mirror strip , Fresnel lens, flat plate with adjustable mirrors and compound parabolic concentrator)- non concentrating types (Flat plate and evacuated tubes)-Comparison between concentrating type and non concentrating type.

Solar Thermal storagesystems: Basic concept of Sensible heat storage-Latent heat storage-solar pond.

CO2	Explain the various technologies for solar heating systems.		
M2.01	Illustrate the utilisation of solar energy in domestic applications.	4	Understanding
M2.02	Explain the utilisation of solar energy in industrial applications.	4	Understanding
M2.03	Illustrate the solar space heating and space cooling.	3	Understanding
M2.04	Classify solar power plants based on temperature.	4	Understanding
	Series Test – I	1	

Contents :

Domestic applications - Solar thermal devices-Construction-working -merits and demerits -Solar cooker(box type and dish type)-solar water heater(Natural circulation & forced circulation) –solar domestic water heating system

Industrial applications -Schematic arrangement and operation -solar furnace-Solar still (single slope & double slope)-solar pumping -solar air heater- solar crop dryer –solar kilns-solar wax melter-solar pump-Solar green houses.

Solar space heating- Basic concept with sketches- passive techniques (direct gain, indirect gain and isolated gain) -active techniques (hot air solar heating system and solar steam heating system).

Solar Space cooling-Block diagram and working of solar absorption technique.

Solar Power plants-Classification of solar power plants based on Temperature (low/medium/high)- Schematic diagram - working -Low temperature solar power

plant(Flat plate collector power plant)-Medium temperature solar power plant (Line focusing parabolic collector power plant)-High temperature solar power plant (Central receiver power plant).			
CO3	Identify different interconnections of Solar Photovoltaic (SPV) modules.		
M3.01	Draw the V/I characteristics and choose various parameters of a solar cell	4	Applying
M3.02	Illustrate various array configurations of solar cells.	4	Understanding
M3.03	Choose the SPV modules and their connections to meet a specific power requirement.	4	Applying
M3.04	Explain various types of solar PV systems.	3	Understanding
Contents : **Solar cells:** Working principle (photovoltaic effect)- Basic construction -merits and demerits -Classification of Solar Cells (On the basis of thickness of active material, junction structure and type of active material) **V/I characteristics of solar cell**-Definitions -short circuit current (Isc) -open circuit voltage (Voc) -Maximum Power (Pmax)-voltage at maximum power (Vmp)- current at maximum power (Imp)- fill factor-conversion efficiency-simple problems -equivalent circuit of a solar cell. **Array Configuration**:Definitions-cells-module-panel-array -Basic interconnections of solar cells (series, parallel and series parallel) -solar PV string (single string in series systems -multiple string in series systems)-V/I characteristics of series, parallel and series parallel combinations **Solar photovoltaic (PV) module**: Rating of PV module-Number of cells in a module-Estimating Wattage of PV module-Estimating number of PV modules -Total power -simple problems **Types of solar PV systems:**-Stand alone systems, Grid connected systems, Hybrid SPV systems (Block diagram and description).			
CO4	Identify various components in Balance Of Systems (BOS) for PV installation.		
M4.01	Name the BOS components for Solar Photovoltaic systems.	2	Remembering
M4.02	Select batteries for SPV systems.	4	Applying
M4.03	Illustrate the working of solar power converters and controllers.	4	Understanding

M4.04	Outline the applications of Solar PV Systems.	4	Understanding
	Series Test – II	1	

Contents :

Balance of Systems(BOS)- Concept - BOS components-(List of items and their purpose only)

Batteries- Battery for PV systems-Choice of battery-Estimating Number of Batteries required in parallel and series connections-Simple problems

Solar Power converters and controllers -power converters- basic concept with diagram (Buck converter ,Boost converter)-charge controller(block diagram approach) -MPPT-PWM

Solar PV applications:-Block diagram approach-Water Pumping-Lighting-Medical Refrigeration-Village power-Telecommunication and signaling.

Text / Reference

T/R	Book Title/Author
T1	Solar Energy: Fundamentals and Applications; H. P. Garg & J. Prakash; 2000; Tata McGraw-Hill.
T2	Solar Photovoltaic Technology and Systems: A Manual for Technicians, Trainers and Engineers Chetan Singh Solanki-PHI learning Pvt Ltd
R1	Non-conventional Energy Resources-B H Khan Mc. Graw Hill Education
R2	Solar Photovoltaic: Fundamental, Technologies and Applications; C.S. Solanki; 2011; Prentice Hall of India.
R3	Solar Energy: S P Sukhatme,J K Nayak- Mc. Graw Hill Education
R4	Solar Energy Utilisation, G. D. Rai, Khanna Publishers
R5	Handbook of Photovoltaic Science and Engineering; Antonio Luque, Steven Hegedus; 2003; John Wiley and Sons.

Online Resources

Sl.No	Website Link
1	NPTEL :: Physics - NOC:Solar Energy Engineering and Technology https://nptel.ac.in/courses/115/103/115103123
2	https://nptel.ac.in/courses/103/103/103103206
3	https://nptel.ac.in/courses/112/105/112105050
4	https://nptel.ac.in/courses/112/105/112105051
5	https://nptel.ac.in/courses/108/105/108105153
6	https://nptel.ac.in/courses/112/104/112104300
7	https://nptel.ac.in/courses/115/107/115107116
8	Solar Thermal Technologies - energypedia https://energypedia.info/wiki/Solar_Thermal_Technologies
9	Solar Heating and Cooling Technologies | US EPA https://www.epa.gov/rhc/solar-heating-and-cooling-technologies
10	Handbook_for_Solar_PV_Systems.pdf (bca.gov.sg) https://www.bca.gov.sg/publications/others/handbook_for_solar_pv_systems.pdf